MSc Big Data Analytics and Decision Making

Module Code: MOD009771
Assessment: Individual Big Data Analytics Project (100%)
Word Limit Equivalent: 3,000 words
Submission Deadline: 10 April 2026 at 14:00

Assessment Scenario

You are working as a Data Analytics Consultant for an e-commerce company. The company wants to use historical transaction data to improve customer understanding, sales performance, and decision-making.

You will use a real public dataset:

Dataset: Brazilian E-Commerce Public Dataset by Olist (Kaggle – Free)

This dataset contains real-world online retail data including customers, orders, products, and payments. The data is not ready for analysis and requires cleaning, integration, and modelling.

Datasets to Use

Core Files (Required)

1. olist_orders_dataset.csv
2. olist_order_items_dataset.csv
3. olist_order_payments_dataset.csv
4. olist_customers_dataset.csv
5. olist_products_dataset.csv

Optional (For deeper analysis / higher marks)

- olist_order_reviews_dataset.csv
- olist_sellers_dataset.csv
- product_category_name_translation.csv

PART 1 — Data Acquisition, Integration & Preparation (30%)

You must design and implement a structured data preparation pipeline using Python.

The aim of this section is to demonstrate your ability to understand, connect, clean, and prepare multiple related datasets for analytical use.

Task 1. Data Integration

You must:

- Use the core datasets provided to build a logically connected analytical data structure.
- Identify and demonstrate how the datasets relate to one another using keys such as:
    - order_id
    - customer_id
    - product_id
- Clearly explain the relationships between tables (e.g., one-to-many relationships between orders and order items)

Important:
You are not required to combine all datasets into one single flat file. Instead, you should preserve the relational structure and show how tables are linked. You may create merged or summary datasets *only where needed* for analysis.

You should also justify whether any optional datasets were included or excluded and explain why.

Task 2. Data Quality Evaluation

You must **critically evaluate and provide evidence of**:

- Missing values

- Inconsistent formats (e.g., dates, text fields, IDs)
- Duplicate or redundant records
- Structural/Data issues arising from multi-table relationships

You must assess the impact of these issues on business decision-making.

Task 3. Data Cleaning & Feature Engineering

You must:

- Apply appropriate and justified data cleaning techniques
- Standardise formats and resolve inconsistencies.

You are also required to engineer new analytical variables derived from the existing data fields. These variables are not directly available in the dataset and must be calculated by combining information from multiple tables.

Examples include:

Analytical Variable	**How It Can Be Derived**
Delivery delay	Difference between *actual delivery date* and *estimated delivery date* in the orders dataset
Customer lifetime value (CLV)	Total value of all purchases made by a customer (sum of order payments or order item values)
Purchase frequency	Number of orders placed by each customer
Average basket value	Average monetary value per order (sum of items per order)
Revenue per seller	Total sales value associated with each seller from order items

NOTE: You may create additional meaningful variables where relevant.

All engineered variables must be:

- **Technically correct** (clearly showing how they were calculated)

- **Business-justified** (explaining why they are useful for decision-making)

Deliverable for Part 1 (Max 1,000 words)

Your report must:

- Critically evaluate data quality issues identified
- Justify integration and cleaning decisions
- Include annotated Python code snippets that clearly show how new analytical variables were derived.
- Present summary statistics before and after cleaning

PART 2 — Relational Database Implementation & Business Analytics (SQLite) (50%)

You must implement a relational database using SQLite via DB Browser for SQLite and answer some business questions using SQL.

Task 1: Database Design (Schema + Data Integrity)

You must:

- Create a database file named: olist_analysis.db
- Design an appropriate schema using the cleaned core datasets from Kaggle (and optional datasets if used).
- Define primary keys and foreign keys where appropriate.
- Explain and justify key design decisions (e.g., how you manage one-to-many relationships, choice of keys, handling of optional tables).

Minimum expected tables: customers, orders, order_items, payments, products.
(Optional: sellers, reviews, category translation)

Task 2: Data Import into SQLite

You must:

- Import your cleaned datasets (preferred) into SQLite tables
- Verify import success:
    - record counts per table
    - key fields populated as expected
- Demonstrate that your schema supports reliable joins

Evidence required:

- Screenshots of table structures.
- Screenshots of row counts (or SQL count queries).

Task 3: SQL Business Analytics Queries (Aggregation Required)

You must write SQL queries to answer the following areas. Your queries must use appropriate SQL constructs such as JOIN, GROUP BY, aggregate functions, ORDER BY, HAVING, and date grouping.

A) Customer Insights

- Identify the top 10% highest-spending customers.
- Compute the average order value.

B) Sales Performance

- Identify product categories generating the highest revenue.
- Analyse monthly sales trends (revenue over time).

C) Operational Insight

- Compute average delivery time/delay.
- Identify patterns in late deliveries (e.g., by month, by location, by category).

For each query, you must:

- Provide the SQL statement.
- Provide a screenshot of the output.

- Provide a concise business interpretation (what it means and what action could be taken).

Deliverable for Part 2 (Max 1,500 words within the 3,000 total)

Include:

- Schema design explanation + keys/relationships
- SQL table creation evidence (or exported schema).
- Import evidence (tables + counts).
- SQL queries + screenshots.
- Business interpretation of results.

PART 3 — Business Dashboard (20%)

Create a Power BI dashboard for managers.

Your dashboard must include:

✓ Monthly sales trend
✓ Revenue by product category
✓ Customer value segment (high vs low spenders)
✓ Delivery performance overview

The dashboard must be interactive and easy to understand.

Deliverable for Part 3 (Max 500 words)

Explain:

- Who the dashboard is for
- Why you chose these visuals
- What business decisions it supports

Report Format and Submission Standards

You must submit ONE structured report document that contains all three parts of the assessment.

Report Structure

Your report must be organised using clear section headings:

1. **Part 1 – Data Acquisition, Integration & Preparation**
2. **Part 2 – MongoDB Database & Business Analytics**
3. **Part 3 – Business Dashboard**

Each section should clearly correspond to the tasks outlined in the assignment brief.

Screenshots and Evidence

All screenshots must:

- Be clearly labelled.
- Follow a consistent naming convention.
- Be referenced and explained within the report

Example naming format:

- *Figure 1: Missing Values Identified in Orders Dataset*
- *Figure 2: MongoDB Aggregation Query – Monthly Sales Trend*
- *Figure 3: Power BI Dashboard – Customer Segmentation View*

Screenshots must support your explanation — they should not appear without commentary.

Referencing Requirements

All academic and technical sources must follow **Harvard referencing style**, including:

- In-text citations
- A reference list at the end of the document

You should reference sources such as:

- Documentation (e.g., MongoDB, Python libraries, Power BI)
- Academic concepts (e.g., data cleaning techniques, database design principles)
- Data visualisation best practices.

Report Quality Expectations

Your report must demonstrate:

- Clear academic writing
- Logical structure and flow
- Professional presentation
- Appropriate technical depth for Level 7 study

Poor structure, lack of explanation, or missing evidence may limit marks even if technical work is correct.

Submission Format

Submit **one ZIP file** containing:

- The **single combined report document** (PDF or Word)
- Python scripts
- MongoDB scripts/queries
- Database export or GitHub repository link
- Power BI file or dashboard link

Marking Focus

Area	What We Look For
Data Preparation	Critical evaluation and justified preparation

Database Design	Logical structure and correct use of MongoDB
Queries	Accuracy and business relevance
Dashboard	Clarity and usefulness
Analysis	Ability to interpret and communicate insights